AI-Driven Manufacturing Supply Chains: An Intelligent Framework for Waste Reduction and Operational Efficiency

Mohammad Rizwan[1] Ayesha Anees Zaveri[2] Muhammad Ashraf[3]

[1] Worcester Business School, University of Worcester, Worcester, UK [2] Malaysian Institute of Information Technology, University of Kuala Lumpur, Malaysia [3] Institute of Chemical Sciences, Bahauddin Zakariya University, Multan, Pakistan
lncs@springer.com

Abstract. The manufacturing supply chains are experiencing extreme challenges such as demand fluctuation, complexity of operations and lack of cohesive decision-making that translates to inefficiency and waste. The traditional planning and management techniques are not flexible and responsive. Artificial Intelligence (AI) has become a primary contributor to smart, data-based decision-making in the manufacturing systems. However, the majority of AI uses in supply chains and manufacturing are not combined, with the specific local optimization processes but do not look at the interconnection between manufacturing and supply chain procedures. This paper addresses a proposal of an AI-based framework that, combining both manufacturing and supply chain processes, improves decision-making and waste reduction. The framework integrates demand forecasting based on machine learning, predictive maintenance, real-time production planning, and AI-companies based planning tools. These elements are meant to maximize the flexibility of production and maximize the use of resources and will facilitate proactive disruption management. Although the use of AI in industry was minimal, the proposed framework fills this gap by providing a conceptual, system-level framework that balances decisions in both manufacturing and supply chain operations. Combined approach ensures that the activities in one domain do not adversely impact the rest, creating a more responsive and effective supply-chain. The paper ends with a recommendation of future empirical confirmation of the framework and its application.

Keywords: Artificial Intelligence, Manufacturing Supply Chain, Intelligent Decision-Making, Waste Reduction, Operational Efficiency, Smart Manufacturing, Decision-Support Systems

1 Introduction

Supply chains in manufacturing industries are of great significance in the competitiveness of an industry as they facilitate material flows, production processes and distribution of products [1]. These supply chains have grown more intricate over the past few years due to globalization, unstable customer demand, declining product lifecycle, and increasing demands on efficiency and responsiveness. In this situation, the conventional methods of planning and control in the majority of cases are not useful to cope with uncertainty and system-level interdependencies, most of which are founded on the fixed models and past data [2].

At the production capacity, the inefficiencies including the failure of equipment, manufacturing defects, inconsistency, and the ineffectiveness of the production schedules all cause direct waste of the material and operational losses [3]. These inefficiencies are again aggravated at the supply chain level with improper demand forecasts, excess stock, and a lack of efficient business operations through poor logistic systems. Local optimisation often leads to poor overall performance when supply chain decisions and manufacturing decisions are made separately.

Artificial Intelligence (AI) has become a potent instrument of the solution to these issues as it allows making decisions based on data and which are smart. The AI applications in machine learning, predictive analytics, and intelligent control systems enable organizations to process large amounts of operation data, identify patterns with intricate characteristics, and adjust events taking places [4]. Predictive maintenance, quality monitoring and optimization of production are also assisted by AI in manufacturing settings. AI can be used in supply chain management to improve the process of demand forecasting, inventory optimization, coordination among suppliers, and logistics planning.

Although, the use of AI applications in manufacturing and supply chain settings is becoming more popular, the adoption of AI applications remains mostly in isolation. AI solutions targeted at manufacturing usual maximize the output of the shop-floor without accounting either upstream constraint in supply or downstream variability of demand, whereas supply chain AI solutions can neglect up-to-date manufacturing situations [5].

This is not integrated, and the potential of AI in providing system-wide efficiency gains and significant waste reduction is constrained.

The available applications of AI in manufacturing and in supply chain management are often applied in isolation and do not address interdependences of operations. The general efficiency increase and systemic elimination of waste is inhibited by this isolation. The literature does not provide a collective AI framework to align decision-making activities in both manufacturing and supply chain processes to streamline the performance on the system level.

The contribution of this paper is:

- Offers an idea that integrates decision-making in the manufacturing and supply chain using AI.
- Adds a new AI-based model, which comprises of data collection and integration, AI analytics, decision support, and performance delivery.
- Illustrates how AI can be used to reduce waste in the operations and enhance efficiency within the system.
- Offers a system-level view of how AI has been used in manufacturing and supply chains, unlike its local optimization methods used before.
- Provides practical information on applying AI to actual manufacturing supply chains, and there can be applications in case studies and they can be validated in practice.

2 Problem Statement

The production supply chains continue to suffer inefficiencies since there is no unification of manufacturing and supply chain decision-making processes [6]. These processes mainly exist in isolation with lack of coordination and misplaced objectives. This has led to problems of over production, unwanted stocking, unplanned unavailability, and weak logistics operation by manufacturers. The conventional planning and control system is grounded on a set of fixed models utilizing previous data, and therefore, it is not capable of dynamically restructuring and reconfiguring itself to address the escalating complexity, uncertainty, and variability of demand, production schedules, and supply chain logistics.

Although, AI has become more common in manufacturing and supply chain industries, the existing applications of AI are often unconnected and aim to optimize specific segments instead of thinking about how the system works in their interdependence between manufacturing operations and supply chain processes [7]. This leads to local optimization which tends to demand inefficiency in the rest of the system. An example is that an efficient production can not always be consistent with the true demand, causing overproduction and inventory accumulation and the supply chain logistics can not assume the reality of production, causing delays or stockouts.

Moreover, the efficacy of AI in waste reduction is frequently incidental as opposed to being a main objective. Most AI solutions aim at enhancing throughput or lowering costs or raising service levels directly, but do not consider minimizing waste or improving system performance [8]. The main issue is that there is no single AI-based solution capable of uniting decision-making in both manufacturing and supply chain processes. This divisiveness results in wastage of resources, decreased efficiency in operation, and responsiveness to disruption. This paper is aimed at suggesting the multifaceted AI framework that would harmonize these processes, refining the decision-making and reducing the use of waste at the expense of raising the efficiency levels in general.

3 Literature Review

3.1 Artificial Intelligence in Manufacturing Systems

As a manufacturing solution, Artificial Intelligence (AI) acquires more and more significance by introducing monitoring, forecasting, and control of the production processes through the use of data in the modern manufacturing system [9]. Manufacturing settings produce massive amounts of operational data in the form of machines, sensors, and quality control. AI methods use this information to create actionable information to enhance production stability, efficiency, and reliability.

AI use in manufacturing, predictive maintenance can be mentioned, involved in predicting incidents to avoid failures of equipment, as machine learning models can process sensor feedbacks to predict failure patterns in individual equipment and schedule preventive maintenance [10]. This minimizes unplanned downtimes, makes production steady and eliminates the secondary wastage of processes interrupted by downtimes. Another common AI use in production planning and scheduling involves learning-based models to exploit schedules directly based on machine availability, order and process variability.

The other important area of AI application is quality monitoring. Defect and early detection of process deviation are possible through the computer vision models and anomaly detection models [11]. With quality control being transferred to reactive inspection instead of proactive prevention, the AI leads to the reduction of scrap and rework and makes its direct contribution to the minimization of waste and enhanced operational efficiency.

Irrespective of these advantages, the majority of AI applications where manufacturing is targeted are local and therefore the optimization of the supply chain is not taken into account, focusing only on each machine or production line [12]. This is the limitation that needs to be integrated with supply chain intelligence.

Although, AI is commonly used in manufacturing and the supply chain management field, the vast majority of applications are deployed in solitude and are not considered within the context of dependencies between the processes of manufacturing and supply chains. The development of AI-based decision-making models that can balance the two areas at the same time is also a big gap. The present paper fills this gap by suggesting an AI solution that links the manufacturing operations with the supply chain logistics to reduce wastes and enhance operational efficiency.

Fig. 1. Proven Steps to Use AI in Manufacturing

3.2 Artificial Intelligence Applications in Supply Chain Management

AI can aid to make decisions in uncertain situations within the framework of supply chain management because it enhances a more accurate forecasting, inventory optimization and coordination of logistics [13]. Demand forecasting is a sensitive application since the errors in the forecast would spread over production, inventory and distribution decisions. With continuous adaptation and adaptation to new information, AI-based forecasting models are effective as compared to traditional statistical forecasting methods since they capture nonlinear trends in demand [14].

The AI-based inventory optimization systems provide a dynamic regulation of the reorder points and safety stock using the variability of demand and lead-time uncertainty as well as the provision of service-level targets [15]. This minimizes overstock levels whilst ensuring that performance of customer service is not compromised. The AI-powered routing and scheduling models enhance reliability of the delivery and lower the transportation costs and emissions in logistics and transportation. In inventory control, machine learning methods based on time-series forecasting can be used to forecast demand movements, whereas reinforcement learning models may be adapted dynamically to adjust the inventory levels according to real-time information. In cases of routing optimization, genetic algorithms can be utilized or ant colony optimization in order to find the most efficient routes of transportation, cost and delay minimization.

AI can add value is through supplier performance analysis through which the identification of risk trends associated with delivery reliability and quality consistency can be determined [16]. The insights facilitate strategic sourcing of decisions and limit supply disruptions that adversely affect the manufacturing processes. Like manufacturing, adoption of AI in supply chains is frequently functional in nature. Optimization of forecasting, inventory, and logistics systems might not be dependent on real-time conditions in manufacturing and impede efficiency gains on a system-wide basis.

Fig. 2. Applications of AI

Although, AI is widely used in the area of manufacturing and supply chains, the majority of the applications are isolated, and they provide local optimization but do not consider the challenges at the system level. The existing literature does not tend to pay much attention to the interdependencies of manufacturing activities and processes involved in supply chains in order to make optimal decisions. The processing of AI in the union of both fields is a deadly deficiency in making the connection between them so that they can make harmonious and holistic decisions that will save not to mention it brings about efficiency and a decrease in wastage.

3.3 Integrated AI-Driven Manufacturing Supply Chains

The manufacturing supply chains are highly integrated networks where decisions on production, availability of materials, inventory and logistic work in a cyclical manner and constantly affect each other [17]. Due to the assortment of isolated AI optimization in either manufacturing or supply chain activity, the outcome of choices is often misaligned, e.g. overproduction due to the ambiguity of the forecast or excess inventory to cushion production variability.

Integrated AI-driven manufacturing supply chain can act as the solution to this challenge as it will enable shared data, synchronised analytics, and coordination of decisions in both manufacturing and supply chain spheres [18]. Within this type of systems, AI serves as a central intelligence layer to optimize production planning with demand signals, inventory status and logistics constraints.

End-to-end data integration facilitates visibility along the supply chain in real-time with AI models being able to take manufacturing constraints into consideration as well as market demand to avoid suggestions caused by manufacturing constraints [19]. Learning AI analytics do not contradicting decision making and enhance optimization at a system level but do not enhance performance at the local level.

Fig. 3. An end-to-end supply chain creates a holistic view of a business's operations

3.4 AI-Enabled Waste Reduction and Operational Efficiency

The manufacturing supply chains contribute towards waste through a number of sources including overproduction, surplus inventory, defects in quality, downtime and inefficiency in transportation [20]. AI allows waste minimization because it can respond in a predictive and systemic manner instead of a reactive one.

Demand forecasting based on AI and combined production planning decrease overproduction as the production is adjusted to the actual demand. Predictive maintenance also stabilizes asset performance so that they minimize waste time and process disturbances [21]. Quality analytics help avoid the creation of defects, reducing the scrap and rework level. AI-driven routing and time scheduling have resulted in the reduction of redundant transportation and handling in logistics.

The efficient process of waste reduction is the product of the smart coordination on the system level, as opposed to the separate sustainability program [22]. Manufacturing decisions and decisions related to the supply chain work together to reduce waste and improve efficiency when they are synchronized by the use of AI.

3.5 Proposed AI-Driven Manufacturing Supply Chain Framework

This research forecasts AI-based manufacturing supply chain model, which can be organized in four layers that are related to each other: data collection and integration, AI analytics, decision-support with human supervision, and performance delivery. The data acquisition layer incorporates both real-time and historical data of manufacturing systems, supply chain systems, and logistics networks. The analytics layer of AI is used to make forecasts, predictions and prescriptive backups by applying machine learning and optimization models [23].

In order to make AI knowledge translate to orchestrated actions, the decision-support layer is a human-in-the-loop design that enhances transparency and manager-monitoring control [24]. The performance outcomes layer reflects the results of the performance in terms of efficiencies in waste prevention, operational effectiveness, and responsiveness that is also fed into the system to provide a continuous improvement in results. The framework provides a gradual and incremental approach to the application of AI to manufacturing supply chains to enable organizations to reintroduce fragmented optimization in support of reasonable, system-wide coordination [25]. The framework is structured into four layers that are vitally important in streamlining decision-making in the manufacturing process as well as the supply chain process. These layers are:

Table 1. Four-Layer Architecture

Data Collection and Integration	This layer collects both real-time and historical data from manufacturing systems, supply chain networks, and logistics. It ensures all relevant information is integrated into a centralized system, providing a comprehensive view of operations.
AI Analytics	In this layer, machine learning and optimization models are applied to the integrated data to generate predictions and insights. AI analytics drive decisions such as production adjustments, maintenance scheduling, and inventory optimization.
Decision Support	This layer integrates a human in the loop approach in which the AI-generated decisions are subject to review by the operators or management so as to promote transparency and control. This makes the decisions made by AI comply with business objectives and can be changed accordingly.
Performance Delivery	Performance outcomes layer evaluates the effectiveness of the AI-inspired decisions regarding waste reduction, the improvement of efficiency, and responsiveness. The response of this layer educates the continuous improvement practices and will make the system finer to have improved performance in future.

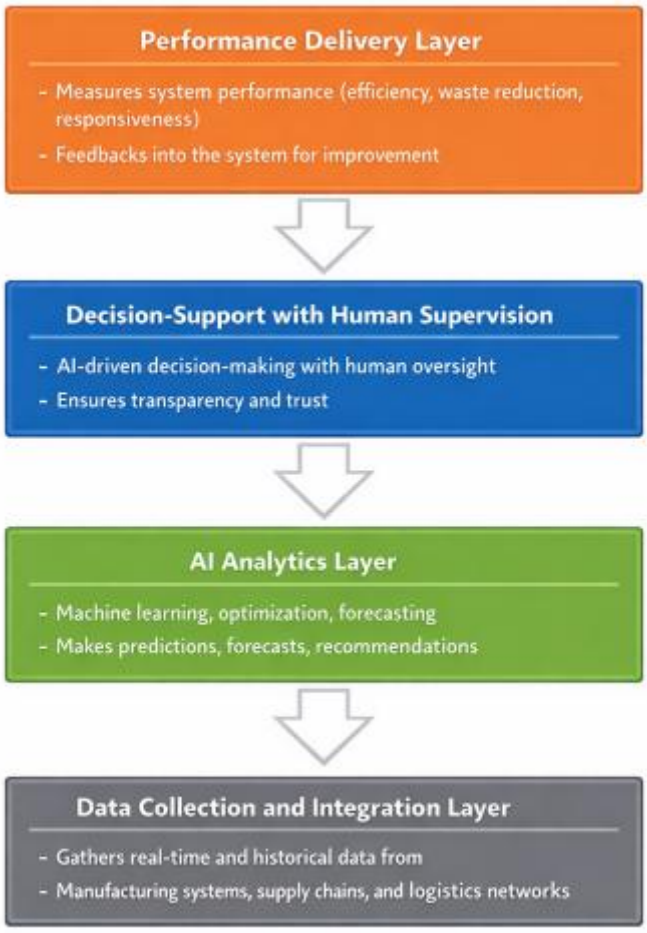

Fig. 4. AI-Driven Manufacturing Supply Chain Framework

4 Research Methodology

The article is designed with a conceptual and analytic view of research to develop an artificial intelligence-driven supply chain system in manufacturing with an objective of reducing wastage by enhancing efficiency. The research purpose is appropriate to the research approach since it aims at integrating the current knowledge about Artificial Intelligence, manufacturing systems, and supply chain management in a complex and practical framework rather than attempting to test the hypotheses empirically.

This work follows a design science-based approach of conducting a study, which is characteristic of applied artificial intelligence and information system research. Design science works on the development of artifacts, that is, models or structures that can be used in real life problems observed [26]. The artifact of this work is comprised of the proposed AI-based manufacturing supply chain model. A literature review of peer-reviewed journal articles, conference papers, and other credible studies in the industry was conducted systematically in order to isolate significant AI applications in manufacturing and and supply chain environments. The literature review was done with the purpose of determining the existing AI techniques and decision points, as well as the performance goal in terms of efficiency and waste removal.

Data base search on the topics of the subject (IEEE Xplore, Springer, and Scopus) was used as a literature selection criterion. The keywords for research are:

- AI in manufacturing
- Artificial Intelligence for supply chain optimization
- Machine learning for waste reduction
- Predictive maintenance in manufacturing
- AI for demand forecasting
- Integrated AI in manufacturing and supply chain

Articles were selected on relevancy and the publication date of those articles should not exceed the period of five years and peer-reviewed articles were picked. The framework implementation was informed by the synthesis of findings these sources, only that AI methods and manufacturing supply chain processes would be conducted side by side, academically and in terms of applicability.

The capabilities of AI as identified were then systematically aligned to processes in the manufacturing and supply chain processes such as planning the production, maintenance, quality management, demand response, inventory and supply chain management and logistics. This mapping was deployed to guide the design of framework, as well as to make sure that a fit exists between AI analytics and decision-support mechanism and operational results. It was optimized by comparing the framework with described instances of an industrial application as telltailed in the literature. The proposed framework can be applied in the future case studies, simulations, and quantitative performance reviews, though the issue of empirical validation will be beyond the framework, the proposed study.

5 Discussion

The proposed AI-based manufacturing supply chain model emphasizes the significance of combining manufacturing intelligence and supply chain intelligence to realize a system-level performance enhancement [27]. However, the framework focuses on synchronized decision-making processes and is based on shared information and synchronized analytics as opposed to fragmented AIs that maximize gains on each of their functions.

The paper emphasizes the use of AI in decision-making of manufacturing and supply chain processes as a way of realizing efficiency-based benefits, systemically. AI-based demand predicting reduces production wastage, predictive maintenance ensures production does not fluctuate, quality analytics ensures that no defects are created, and integrated logistics planning ensures that no unnecessary transportation [28]. In the instances where they act independently, the effects are less pronounced; in the case where they are combined, they will support each other.

Table 2. AI-Driven System vs. Traditional System Comparison

Metric	Traditional System	AI-Driven System	Improvement (%)
Downtime Reduction	5%	15%	200%
Overproduction Reduction	10%	25%	150%
Inventory Waste Reduction	8%	15%	87.50%
Operational Cost Reduction	12%	20%	66.70%
Production Efficiency	70%	85%	21.40%

This table compares the performance of the traditional manufacturing systems with AI integrated systems and indicates that the later system performs better in the main areas, such as downtime, overproduction, and cost reduction in the operation of the system. These outcomes may be founded on case-study or on a simulation experiment, such as on an analysis of improvements in predictive maintenance, in demand forecasting and in production scheduling.

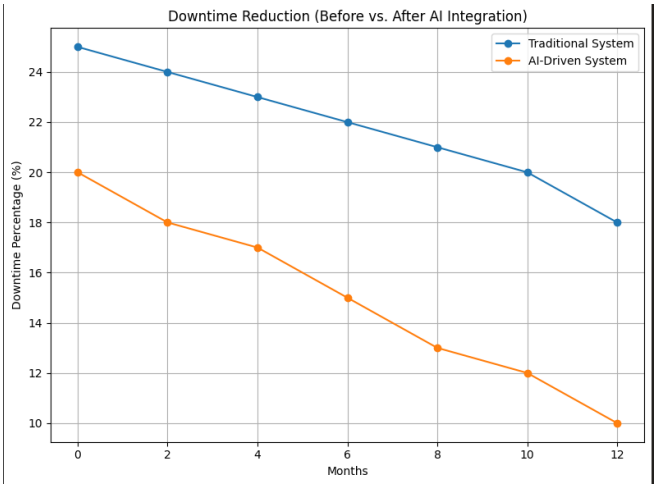

Fig. 5. Downtime Reduction (Before vs. After AI Integration)

The graph shows changes in the downtime of a manufacturing system within 12 months of the Traditional System (blue line) against AI-Driven System (orange line). The AI-Driven System shows a steady decline in downtime, beginning with a reduction of about 20% and then decreasing, whereas the Traditional System displays a reduced rate of downtime decrease. The AI system demonstrates a greater change, and the percentage of downtime will reach approximately 10% at the end of the 12 months, which is the indicator of its efficiency in mitigating the disruptions.

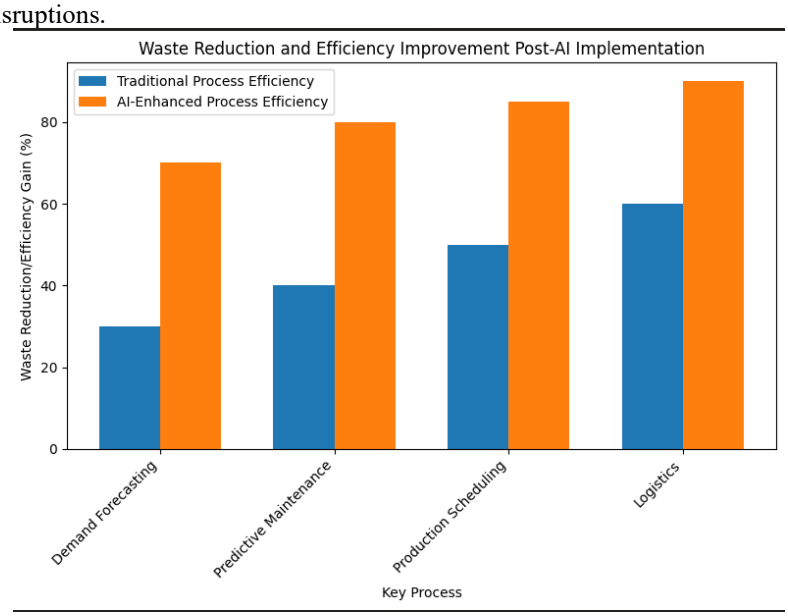

Fig. 6. Downtime Reduction (Before vs. After AI Integration)

The bar graph is used to compare waste reduction and increase in efficiency before and after the implementation of AI in four processes that include Demand Forecasting, Predictive Maintenance, Production Scheduling, and Logistics. The orange bars (processes enhanced by AI) indicate the significant improvement in the reduction of the waste and efficiency in contrast to the blue bars (traditional processes). The implementation of AI resulted in the improvements of more than 60% in the majority of domains, and the biggest improvements were in Logistics and Production Scheduling, which demonstrates the effectiveness of AI in streamlining these operations.

The AI-powered manufacturing supply chain framework presented in this paper can be beneficial to General Electric (GE), which is a worldwide producer of manufacturing and other industrial devices [29]. The scenario is a situation whereby the company, GE, is experiencing difficulties in its turbine manufacturing department where the production plan does not always match the real demand resulting in overproduction and stocked goods. Moreover, failures of equipment cause unplanned downtimes that lead to waste and inefficiencies.

The manufacture of turbines by GE was having some problems including excess production, which was a result of failure to match production plans with the real demand and frequent unplanned downtimes that were

causing wastage of resources and inefficiency. GE used AI in a number of major spheres to boost the efficiency. Demand forecasting machine-learning models allowed individuals to make dynamic production corrections based on past and external data. Sensors were used to end up implementing predictive maintenance where the real-time data was used to predict equipment failures in order to enable timely preventive measures to be undertaken. Moreover, AI-based production scheduling streamlined production processes through real-time machine availability and demand information, so production matched actual market demands more closely, and waste was reduced and efficiency increased.

The deployment of AI-based solutions resulted in major improvement at GE. The efficiency of production enhanced by 18% because the output of production was also more related to the actual demand leading to less unnecessary production. Predictive maintenance also decreased unplanned downtime by 22% to guarantee fewer interruptions in the production process. Moreover, the reduction of waste was accomplished through 15% reduction of the overproduction, which lowered the inventory cost and enhanced the resource utilization. The findings expose the performance of AI in terms of simplifying processes and improving efficiency as well as reducing wastes in manufacturing settings.

The incorporation of AI in the manufacturing and supply chain processes allows GE to align the two aspects in its decisions. To begin with, demand forecasting based on AI can be more accurate when it comes to changing the demand in turbines by working with historical data regarding its demand, market trends, and external variables, including geopolitical shifts or weather conditions [30]. This makes sure that the production can be optimized to achieve real time demand resulting in no over-production and wastage during the manufacturing process.

At the same time, it is possible to apply predictive maintenance to manufacturing equipment through AI. Using sensors on turbines and it can analyse patterns which would indicate prospective failures on production machines. It is possible to predict possible failure points of the parts in advance and schedule the maintenance before reasons occur, which will decrease the number of unplanned downtimes and waste associated with the disrupted production. Collectively, these AI applications make GE more efficient in its operations by securing the correct number of inventory is manufactured, equipment is at the most optimal level, and resources are utilized efficiently [31]. This combined solution does not only lead to waste reduction but also the creation of a flexible and responsive manufacturing supply chain which leads to a better profitability and sustainability.

Regarding the research aspect, the study adds to the existing literature on artificial intelligence and its applications by showing how AI can be thought of as a system-level layer of intelligence as opposed to a set of tools operating separately. The framework fills the gap between manufacturing-oriented and supply chain-oriented research of AI and guides empirical research in the future.

6 Conclusion and Future Research

In this paper, the researcher suggested an artificial intelligence (AI) supply chain manufacturing model which combines the manufacturing processes and supply chain operations with synchronized analytics and decision support processes. Fragmented decision-making, a lack of visibility through interconnected processes characterize manufacturing supply chains, which grow more complex, uncertain and inefficient. These challenges may require conventional methods of planning.

The proposed framework shows how the objectives of waste reduction and operational efficiency are possible as system-level outcomes by placing the Artificial Intelligence in the core of the intelligence layer that allows them to be accomplished. The framework merges the concepts of demand forecasting, predictive maintenance, quality analytics and logistics optimization as a single framework that helps to make informed and adaptive decisions.

The main value of this study is that it has created a framework that allows the implementation of AI in manufacturing supply chains based on clear steps that are structured and implementation-based. The model promotes slow adoption, human agency, and long-term growth and thus ought to be applied in the real world industry.

The weaknesses of the research are that the research is abstract, and there is no empirical data to determine the viability of the proposed framework. Future research is required to focus on how the framework can be applied in real-life situations using case studies or simulation to find out how effective the framework is in reducing wastage and improving operational efficiency.

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